CS & IT ENGINEERING



Algorithms

Dynamic Programming



Lecture No.- 08

By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic

Multi-Stage Graph.

→ Problems

Topics to be Covered



- Topic
- Topic
- Topic

Travelling Salesman Problem (TSP)

PYQs

Tomorrow: 8AM



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW



Topic : Dynamic Programming :(DP)

28%

PYQ

A = "qpqrr"

B = "pqprqrp"

x = length of LCS

y = number of LCS

~~x = 10y?~~

$x + 10y = ?$

Shortcut

A → 5

B → 7

5 ✗

4 ✓

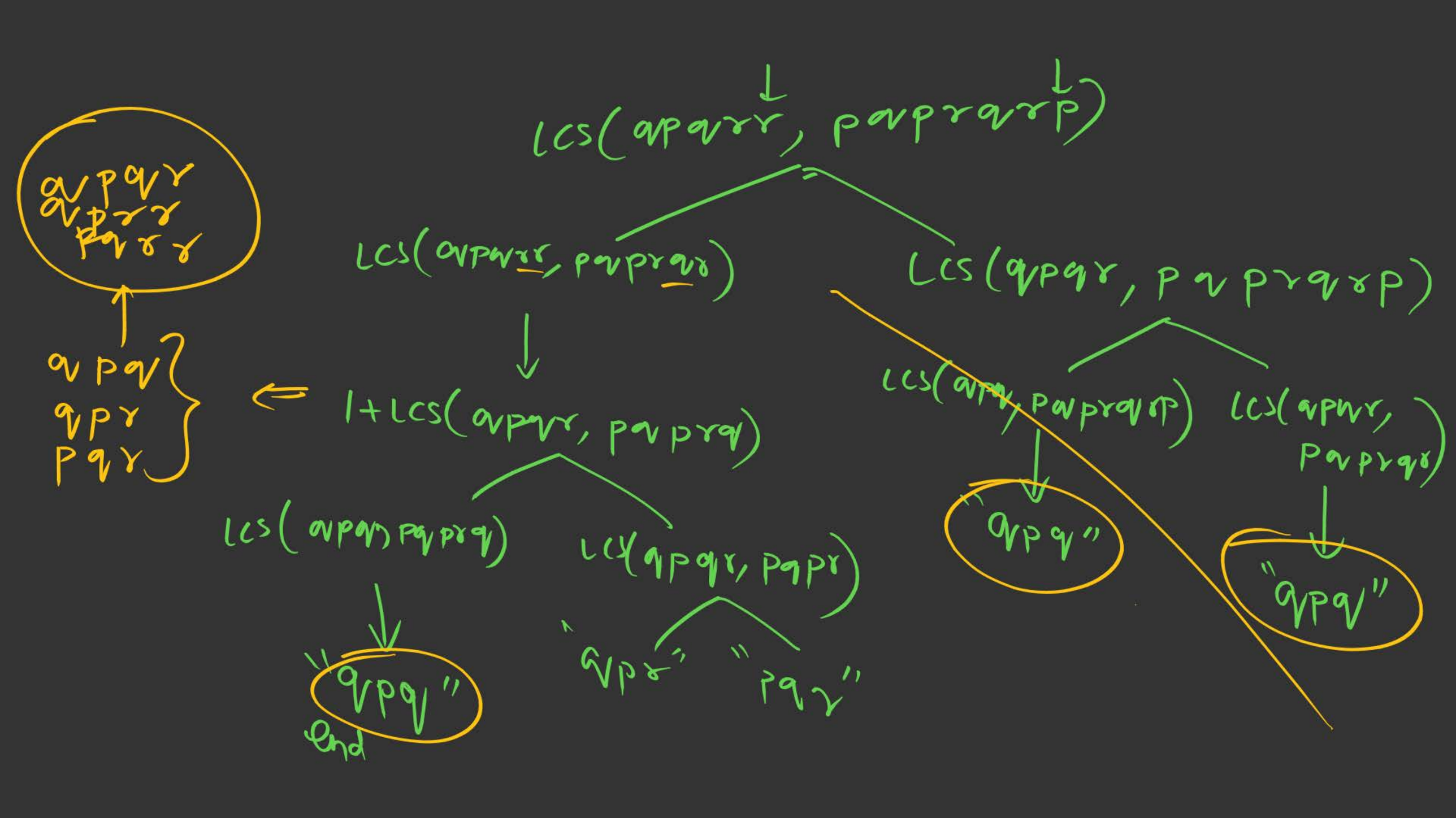
$x = 4$

1) qpqr
2) pqrr
3) qprr

LCS

$y = 3$

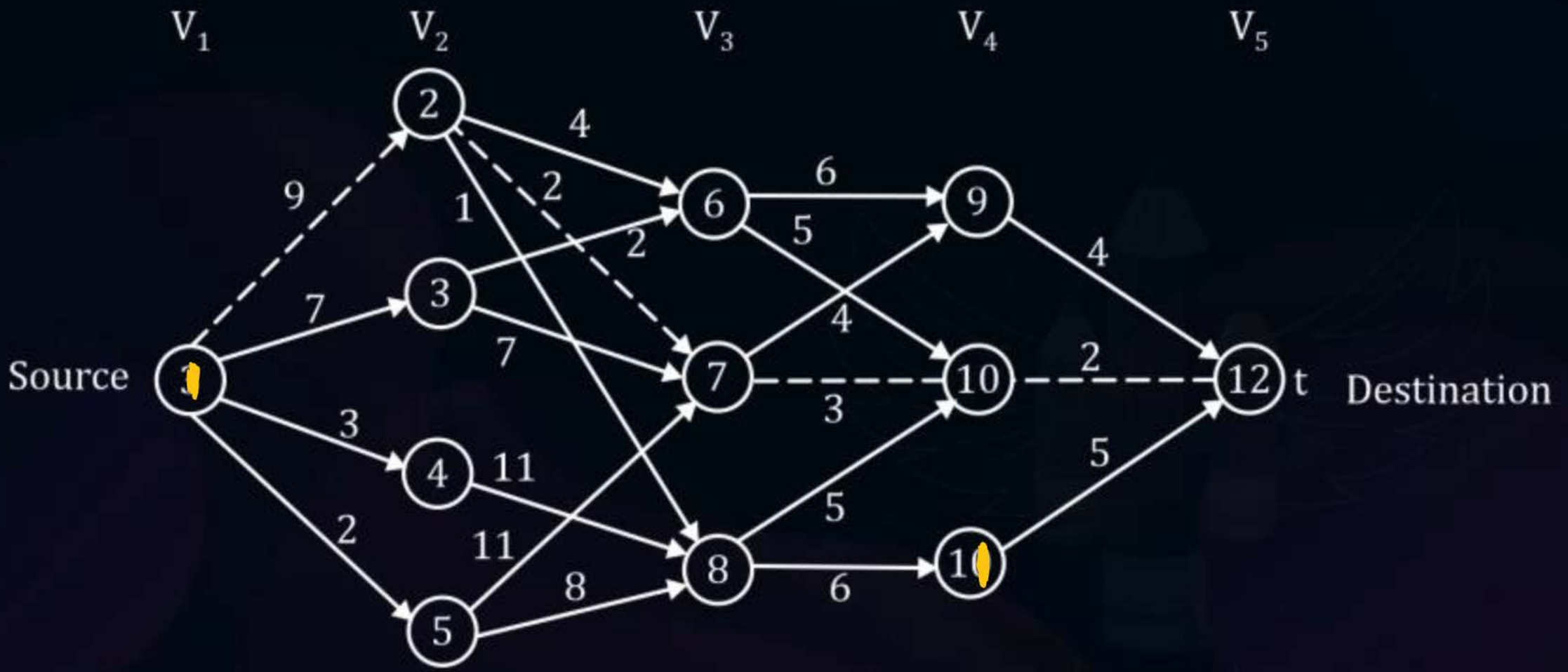
$x + 10y = 4 + 3 \times 10 = 34$


LCS(aabaacc, baabcaca)
LCS(aabaac, baabcac)
LCS(aabaa, baabca)
1+LCS(aaba, baabc)
LCS(aaba, baabc)
LCS(aabac, baabc)
"aaba"
"ac" "bc"
LCS(aabaa, baabcac)
LCS(aaba, baabcac)
LCS(aabaa, baabca)
"aaba"
"aaba"
aabaa
aaba
aba
baa



Topic : Dynamic Programming :(DP)

Multi-stage graph





Topic : Dynamic Programming :(DP)

Multi-stage graph

$$\text{cost}(i, j) = \min \{c[j, k] + \text{cost}(i + 1, k)\}$$

$$= K \in V_{i+1}$$

$$\langle j, k \rangle \in V$$

$l \rightarrow$ stage: $\quad \text{cost}(l - 1, j) = C(j, t)$

$t \rightarrow$ destination $\quad D(I,j) = K \rightarrow$ that k which minimize cost (i,j)

TSP



Topic : Dynamic Programming :(DP)

8. Travelling Salesman Problem (TSP)

The tour of TSP should start from the home city (V_0) and visit remaining (n-1) cities exactly once and comeback to the home city (V_0), such that the cost of tour is minimize.





Topic : Dynamic Programming :(DP)

Home city = Starting vertex (V_0)

Tour: V0 → n-1 (vertices) → V_0

TSP → Optimization problem

Objective function → minimize the tour cost.





Topic : Dynamic Programming :(DP)

Greedy Method

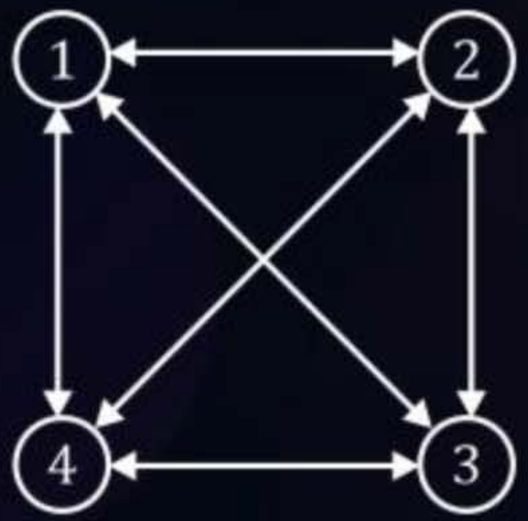
$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$

$= 10 + 9 + 12 + 8$

$= 19 + 20$

$= 39$

$V_0 = 1$



Cost adj. Matrix

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

$3 \rightarrow 2$



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The actual optimal/minimum tour cost is-

$1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 1$

$10 + 10 + 9 + 6$

$= 20 + 15$

$= 35$

Optimal solution

Imp. Note:-

If we solve TSP by greedy, we get min tour cost as 39 $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$

Which is not optimal solution.

Hence, greedy fails for TSP.

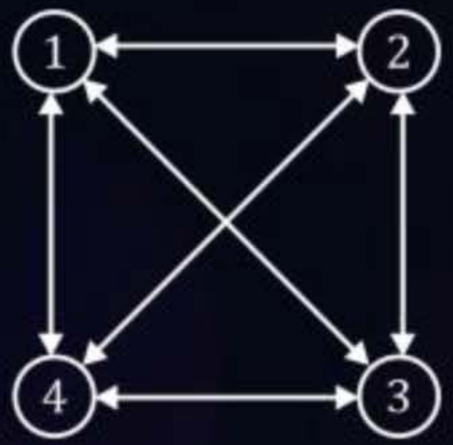
Topic : Dynamic Programming :(DP)

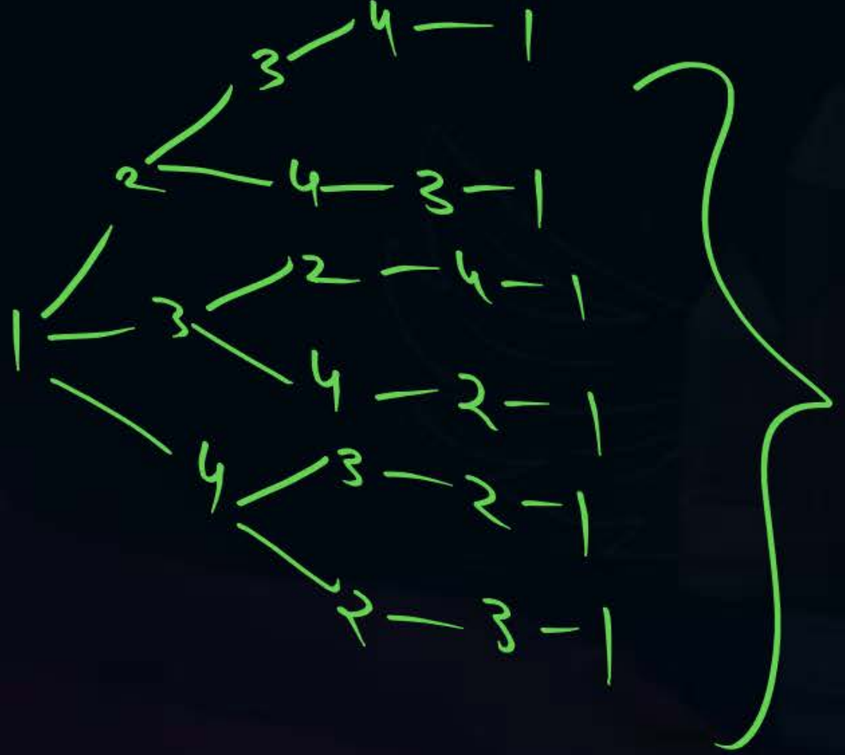
DP based solution: (V.V.imp)

Let g (i, s) represent the cost of the tour of TSP from vertex 'i' visiting are the vertices in the set 's' exactly once and terminating the tour at V_0 (source)

Let, V_0 =1 prev.. E.g.

E.g.





Topic : Dynamic Programming :(DP)

prev graph

Subproblem

$V_0 = 1$

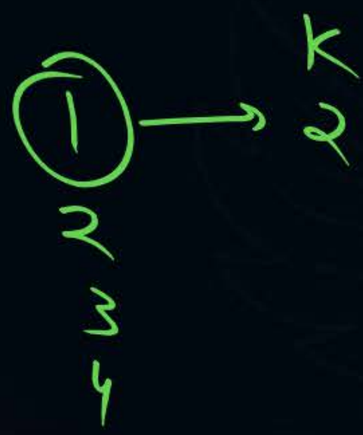
$$g(1, \{2, 3, 4\}) = \min_{K \in \{S\} \text{ and } \langle 1, k \rangle \in E} [C(1, K) + g(K, S - \{k\})]$$

Original Problem → g(1 , s)

[1 → k..... 1] → Sub-problem

C (1, k) + g(K, S –{K})





{2,3,4} - 2

{3,4}



Topic : Dynamic Programming :(DP)

Generalized from → V. imp

V. V. V. V imp

$$g(i,s) = \min [c(i,k) + g(k, s-\{k\})] \ldots\ldots(1)$$

$K \in S$

$\langle i, k \rangle \in E$

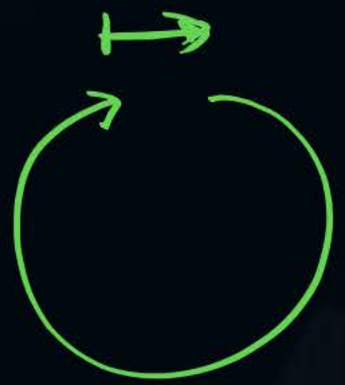
Starting vertex.

$g(i, \phi) = c(i, v_0)$

empty

$i \rightarrow v_0$

To make sure back to source v_0



$J(i, S)$ = Value of k that minimize equation (1)

Topic : Dynamic Programming :(DP)

E.g.



Source (V0)

Cost adj. Matrix

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

Bottom - up

Optimal problem

g(1{2,3,4,}) =

Min
- K =2 : C(1,2) + g(2{3, 4})
- K =3 : C(1,3) + g(3{2, 4})
- K =4 : C(1,4) + g(4{2, 3})

35

g(1,{2,3,4})

Bottom up

Topic : Dynamic Programming :(DP)

1. $|S| = 0$, $S = \phi$

$i = 2, 3, 4$

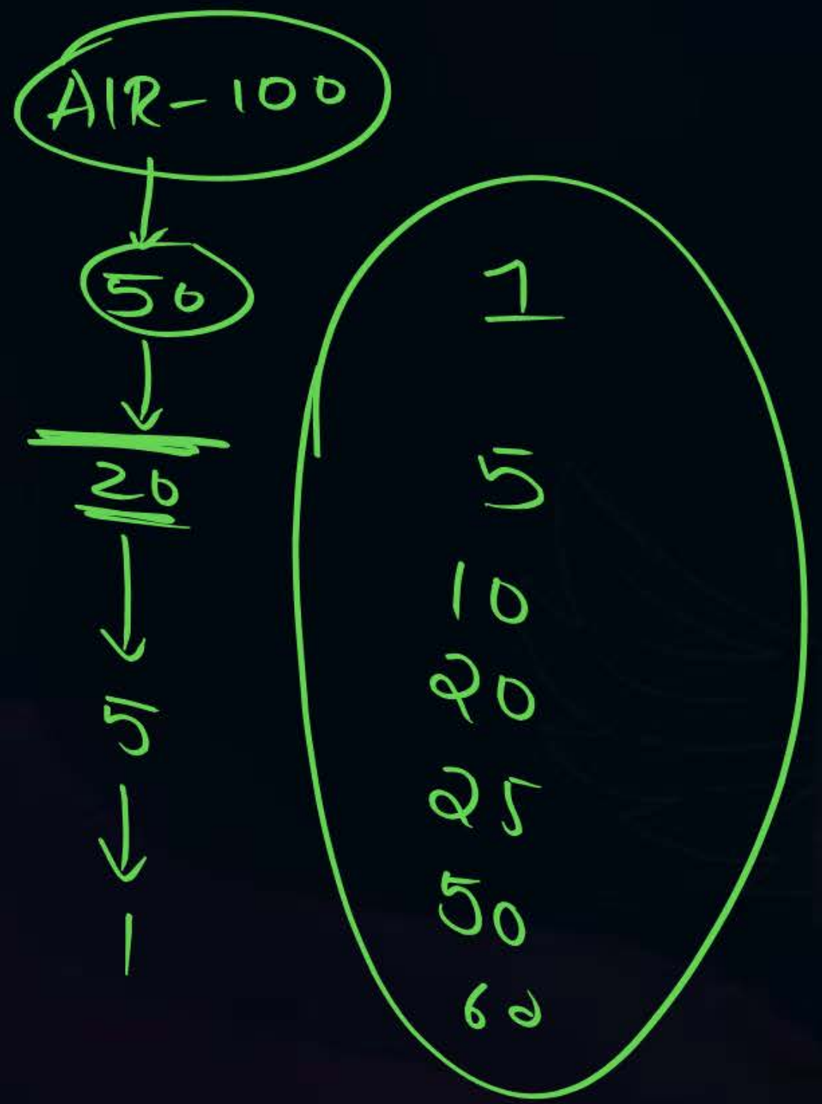
$g(2, \phi) = C(2, 1) = 5$

$g(3, \phi) = C(3,1) = 6$

$g(4\ \phi) = C(4,1) = 8$

$|S| = 0 \quad \phi$

$|S| = 1 \quad 2\ 3$







Topic : Dynamic Programming :(DP)

2. $|S| = 1$

$g(2, \{3\}) = C(2,3) + g(3, \phi)$ ← prev subproblem

$= 9 + 6 = 15$

$J(2, \{3\}) = 3$

$g(2, \{4\}) = C(2,4) + g(4, \phi) = 10 + 8 = 18$

$J(2, \{4\}) = 4$

$g(3, \{2\}) = C(3,2) + g(2, \phi) = 13 + 5 = 18$

$g(3, \{4\}) = C(3,4) + g(4, \phi) = 12 + 8 = 20$

$J(3, \{2\}) = 2$

$J(3, \{4\}) = 4$



Topic : Dynamic Programming :(DP)

$$g(4, \{2\}) = C(4, 2) + g(2, \phi)$$

$$= 8 + 5 = 13$$

$$g(4, \{3\}) = C(4,3) + g\{3, \phi\}$$

$$= 9 + 6 = 15$$

$|S| = 0$ ✓

$|S| = 1$ ✓

$|S| = 2$

$g(4, \{2\}) = 13$

$g(4, \{3\}) = 15$



Topic : Dynamic Programming :(DP)

3. $|S| = 2$

$$g(2, \{3, 4\}) = \min \begin{cases} K = 3: C(2,3) + g(3, \{4\}) \\ K = 4: C(2,4) + g(4, \{3\}) \end{cases}$$

$$= \min \begin{cases} K = 3: 9 + 20 \\ K = 4: 10 + 15 \end{cases}$$

From prev. subproblem

$J(2, \{3,4\}) = 4$

$$= \min \binom{29}{25} = 25$$



Topic : Dynamic Programming :(DP)

$$g(3, \{2, 4\}) = \min \begin{cases} K = 2: & C(3, 2) + g(2, \{4\}) \\ K = 4: & C(3, 4) + g(4, \{2\}) \end{cases}$$

$$= \min \begin{cases} K = 2: & 13 + 18 \\ K = 4 & 12 + 13 \end{cases}$$

$$J(3, \{2, 4\}) = 4$$

$$= \min \begin{pmatrix} 31 \\ 25 \end{pmatrix} = 25$$



Topic : Dynamic Programming :(DP)

$$g(4, \{2,3\}) = \min \begin{cases} K = 2: C(4,2) + g(2, \{3\}) \\ K = 3: C(4,3) + g(3, \{2\}) \end{cases}$$

$$= \min \begin{cases} K = 2: 8 + 15 \\ K = 3: 9 + 18 \end{cases}$$

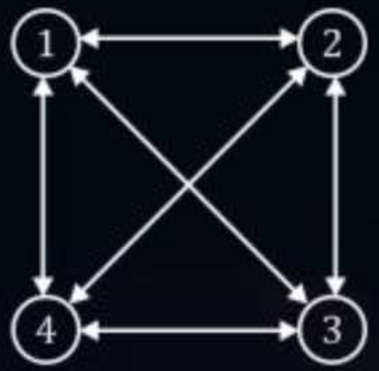
$$J(4, \{2, 3\}) = 2$$

$$= \min \begin{cases} K = 2: 23 \\ K = 3: 27 \end{cases}$$

$$= 23$$

Topic : Dynamic Programming :(DP)

E.g.



Cost adj. Matrix

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

Optimal problem

I (1{2,3,4,}) =

Min
- K =2 : C(1,2) + g(2{3, 4}
- K =3 : C(1,3) + g(3{2, 4}
- K =4 : C(1,4) + g(4{2, 3}

10 + 10 + 6 + 9

= 35

Bottom - up



Topic : Dynamic Programming :(DP)

$$g(1, \{2, 3, 4\}) = \min \begin{cases} K = 2: & C(1,2) + g(2, \{3, 4\}) \\ K = 3: & C(1,3) + g(3, \{2, 4\}) \\ K = 4: & C(1,4) + g(4, \{2, 3\}) \end{cases}$$

$$= \min \begin{cases} K = 2: & 10 + 25 \\ K = 3: & 15 + 25 \\ K = 4: & 20 + 23 \end{cases}$$

$$= \min \begin{cases} K = 2: & 35 \\ K = 3: & 40 \\ K = 4: & 43 \end{cases} = 35$$

$g(1, \{2,3,4\}) = 2$

Topic : Dynamic Programming :(DP)

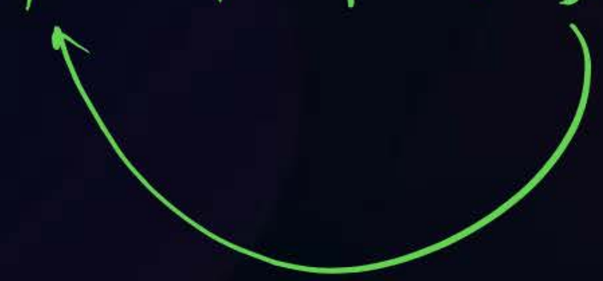
#Q. How to determine the path ~~four~~ for which we got min cost?

tour

$1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 1$

Tour repeated by TSP

Cost = 35



$J(1,\{2,3,4\}) = 2$

$J(2,\{3,4\}) = 4$

$J(4,\{3\}) = 3$

$J(3,\phi) = 1$

Draw Top Down Recursion Tree

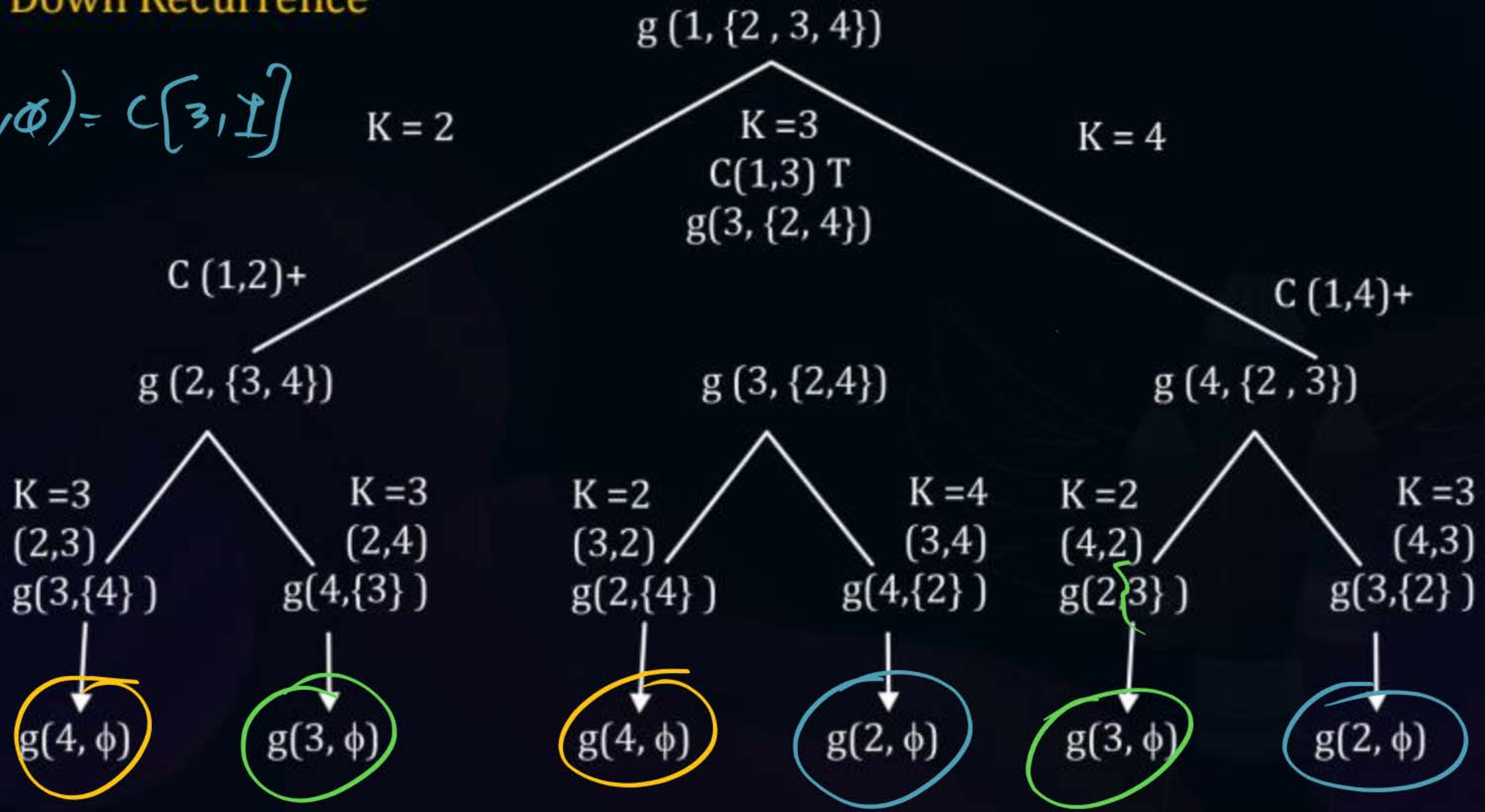
$$g(1, \{2,3,4\})$$



Topic : Dynamic Programming :(DP)

Top- Down Recurrence

$g(3,\phi) = c[3,1]$





Topic : Dynamic Programming :(DP)

Bonus

Held-Karp Algorithm :- DP based Algo to solve the travelling salesman problem (TSP)

1. Time complexity = $O(2^n * n^2)$
2. Space complexity $O(2^n * \sqrt{n})$

n = no. of cities.

(nodes)

Note.: TSP is one of those problems for which there is ~~number~~ NO poly-nominal TC (as of now)

Topic : Dynamic Programming :(DP)

#Q. Dijkstra's SSSP Algorithm when run from vertex A in the below graph, computes the shortest path distance to?

PYQ → 2 mks 40%

Process of elimination

A Only vertex a ✗

B Only vertices a, e, f, g, h ✗

C Only vertices a, b, c, d ✗

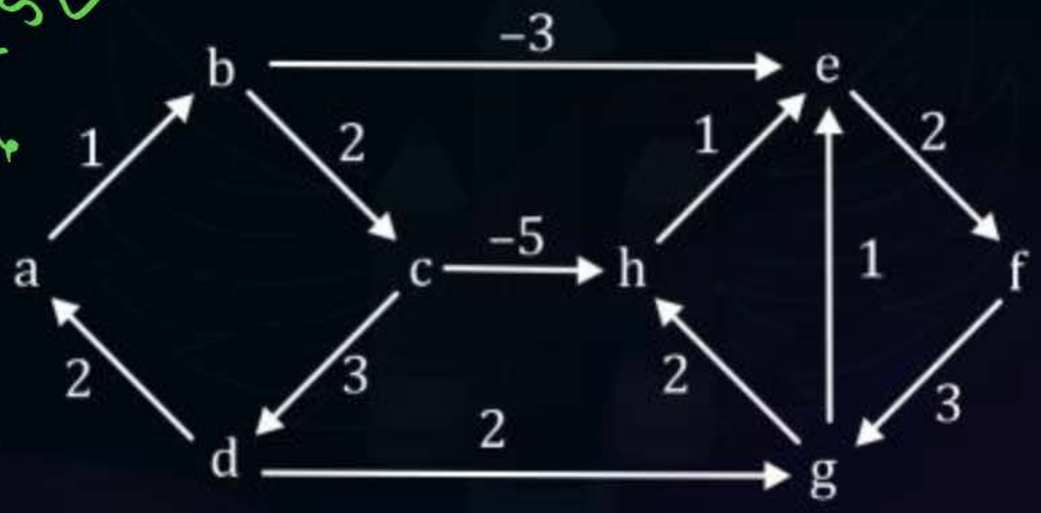
D All the vertices. ✓

-ve wt edges ✓

No -ve wt cycle

A → B ✓

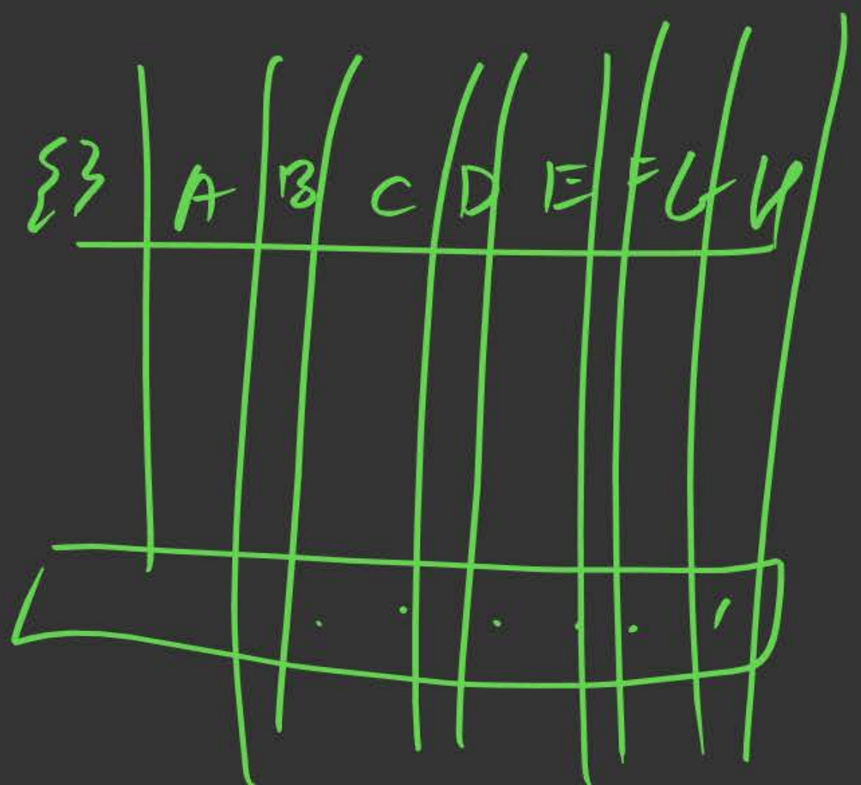
A → E ✓



HW : Solve using

1) Table

2) Spanning Tree



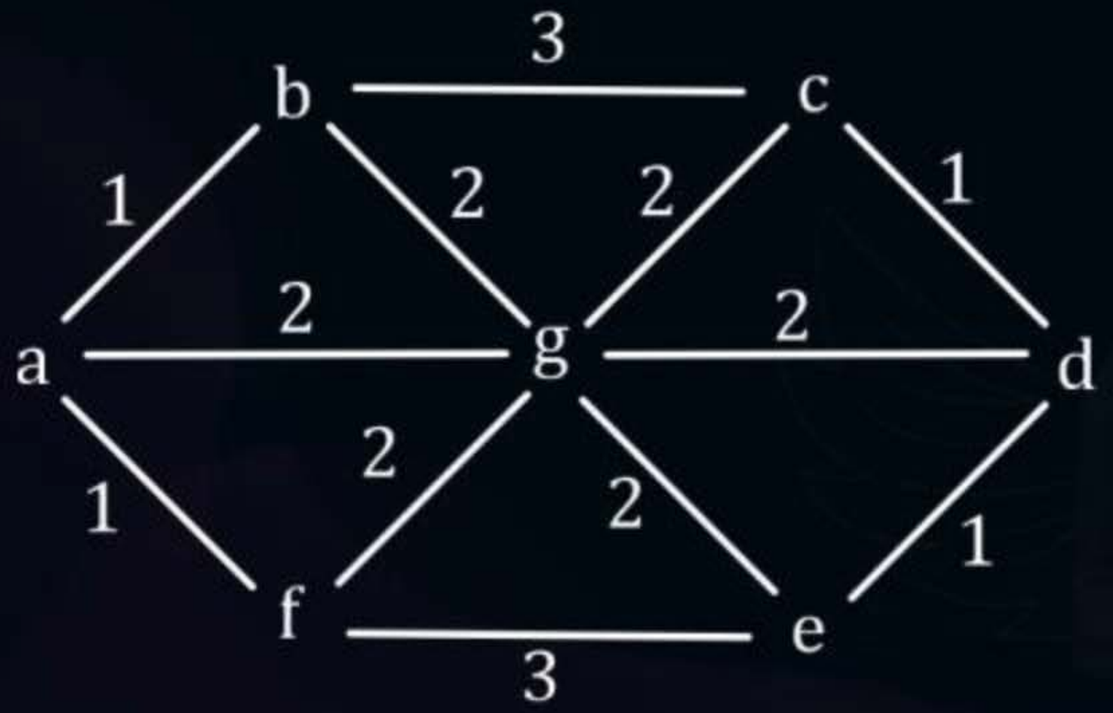


Topic : Dynamic Programming :(DP)

#Q. How many distinct MCSTs?

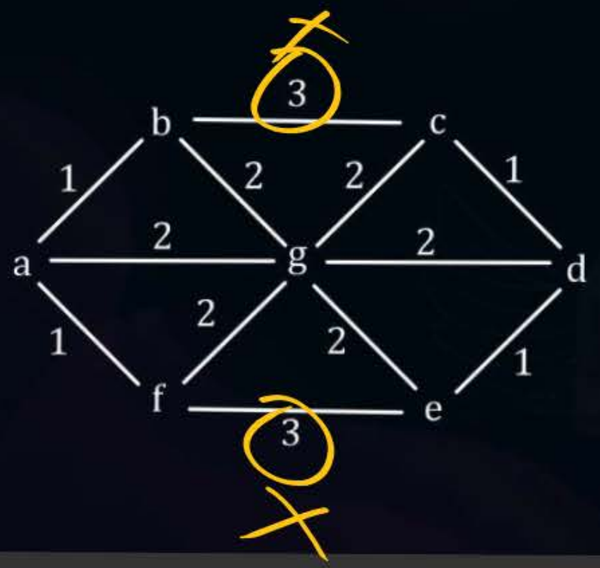
Gate 2024 — 2m

22%

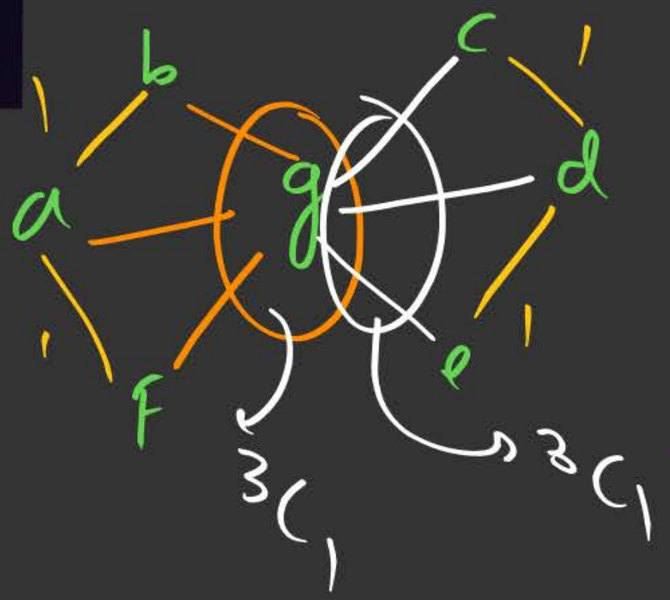


Topic : Dynamic Programming :(DP)

#Q. How many distinct MCSTs?



Kruskal



n = 7
e = 6

$^3C_1 * {}^3C_1$

$= 3 \times 3$

$= 9$ ✓



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

TSP ✓

PYQs ✓



THANK - YOU